上海冠显光电科技有限公司
Shanghai Top Display Optoelectronics Co., LTD

LCD MODULE SPECIFICATION

Customer: _____

Module No.: TS040WVS02CP-B1617A

Date: 2025-01-11

Version: 1.0

For Customer's Acceptance:

Approved by	Comment

Approved by	Checked by	Prepared by
Haitao.Zhou	Chenchen.Wang	Hongwei.Li

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Record of Revision

Rev.	Date	Description	Editor
1.0	2025-01-11	First release	Hongwei.Li

1 General Specifications

No.	Item	Specification	Remark
1	LCD Size	3.95 inch	
2	Driver Element	a-Si TFT active matrix	
3	Resolution	480 (RGB) ×480	
4	Display Mode	Normally Black, Transmissive	
5	Pixel Pitch	0.0487 (H) × 0.1462 (V)	
6	Display Colors	262K	
7	Surface Treatment	--	
8	Color Arrangement	RGB-Stripe	
9	Interface	QSPI/SPI/DSPI	
10	Viewing Direction	ALL	
11	Gray Scale Inversion Direction	--	Note 1
12	Outline Dimension (mm)	84.00(W) × 84.00(H)	
13	Active Area (mm)	71.86 (W) × 70.18 (H)	
14	Touch Screen	CTP	
15	Display Driver IC	TR230S	
16	Touch Driver IC	ICNT8952BN	

Note 1: Viewing direction for best image quality is different from TFT definition. There is a 180°shift.

Note 2: RoHS compliant.

2 Pin Assignment

2.1 Pin assignment

Match connector 1: XF2M-2015-1A（OMRON）or equivalent.

N	Symbol	I/O	Description	Remark
1	GND	P	Ground	
2	VCC	P	POWER INPUT	
3	GND	P	Ground	
4	CS#	I	Chip selection:LOW active	
5	GND	P	Ground	
6	SDO1	I	SPI:SPI-MISO DSPI:SPI-DATA1 QSPI:QSPI-DATA1	
7	SDO0	I	SPI:SPI-MOSI DSPI:SPI-DATA0 QSPI:SPI-DATA0	
8	SCL	I	SPI/DSPI/QSPI:SPI-Clock	
9	SDO3	I	QSPI:QSPI-DATA3	
10	SDO2	I	QSPI:QSPI-DATA2	
11	RESET#	I	TR230S Module_RESET and LCD-RESET: LOW active	
12	WAIT#	I	WAIT# STATUS flag, 0: Sending Command Not Allowed 1: Sending Command Allowed	
13	QSPI-INT	O	Touch interrupt	Note2
14	D/C#	I	DATA or COMMAND flag when choose , QSPI/DUAL SPI: NOT CONNECTION SPI-4WIRE interface: D/C#="H" is DATA or parameter D/C#="L" is COMMAND	
15	TP-INT	O	Touch Interrupt	
16	TP-SDA	I/O	Touch IIC Data signal	
17	TP-SCL	I	Touch IIC Clock signal	
18	TP-RST	I	Touch Reset Signal	
19	TP-VCC	P	Touch Power supply：3.3V	
20	TP-GND	P	Touch Ground	

I---Input, O---Output, P--- Power/Ground

Match connector 2: 2mm spacing 2X7 Pin Header

N	Symbol	I/O	Description	Remark
1	VCC	P	POWER INPUT	
2	GND	P	Ground	
3	GND	P	Ground	
4	CS#	I	Chip selection:LOW active	
5	GND	P	Ground	
6	SDO1	O	SPI:SPI-MISO DSPI:SPI-DATA1 QSPI:QSPI-DATA1	
7	SDO0	I	SPI:SPI-MOSI DSPI:SPI-DATA0 QSPI:SPI-DATA0	
8	SCL	I	SPI/DSPI/QSPI:SPI-Clock	
9	SDO3	I	QSPI:QSPI-DATA3	
10	SDO2	I	QSPI:QSPI-DATA2	
11	RESET#	I	TR230S Module_RESET and LCD-RESET: LOW active	
12	WAIT#	I/O	WAIT# STATUS flag, 0: Sending Command Not Allowed 1: Sending Command Allowed	
13	QSPI-INT	I/O	Touch interrupt	Note2
14	D/C#	I/O	DATA or COMMAND flag when choose , QSPI/DUAL SPI: NOT CONNECTION SPI-4WIRE interface: D/C#="H" is DATA or parameter D/C#="L" is COMMAND	

II---Input, O---Output, P--- Power/Ground

Note1: IM[1:0] Configuration table

IMO	IM1	INTERFACE	Related Pins	Max. Clock
0	0	SPI-4WIRE	CS#、SCL、SDO0、D/C#、WAIT#	100MHz
1	0	QSPI-4SDA	CS#、SCL、SDO0、SDO1、SDO2、SDO3、WAIT#	100MHz

Note2:

The PIN HOST does not need to be connected when the user drives the touch screen directly through the IIC.If the TR230 drives the touch screen, the user needs to detect the PIN to decide whether to read the touch event.

3 Absolute Maximum Ratings

Ta = 25℃

Item	Symbol	Min.	Max.	Unit	Remark
Power Voltage	VCC	-0.30	+5.8	V	
	TP_VCC	-0.50	+3.6	V	
Operating Temperature	Top	-20.0	70.0	℃	
Storage Temperature	T_{st}	-30.0	80.0	℃	
Operating and Storage Humidity	H_{stg}	10%	90%	%(RH)	

4 Electrical Characteristics

4.1 Recommended Operating Condition

VCC=5V， GND=0V， Ta = 25℃

Item		Symbol	Min.	Typ.	Max.	Unit	Remark
Power supply Voltage		VCC	4.5	5	5.5	V	
TP Power		TP_VCC	2.8	3.0	3.3	V	
Input Signal Voltage	Low Level	V_{IL}	-0.3	-	0.3*3.3	V	
	High Level	V_{IH}	0.7*3.3	-	3.3+0.3	V	
Power supply Voltage		I_{VCC}	-	180		mA	VCC=5V, color bar pattern

4.2 Backlight Unit Driving Condition

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Forward Current	I_F	-	40	50	mA	8LEDs (4 LED Serial, 2LED Parallel)Note5
Forward Current Voltage	V_F	-	12.8	14	V	
Backlight Power Consumption	W_{BL}	-	512	700	mW	
Operating Life Time	--	30000	--	--	hrs	Note 2, Note 3

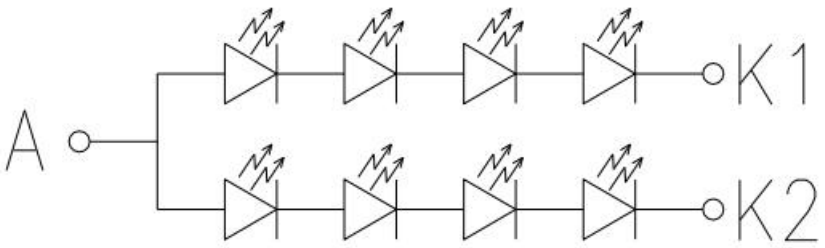
Note1: The LED driving condition is defined for each module (4 LED Serial, 2 LED Parallel).

Note2: When LCM is operated, the stable forward current should be inputted. And forward voltage is for reference only.

Note3: Optical performance should be evaluated at Ta=25℃ When LED is driven at high current, high ambient temperature & humidity condition. The life time of LED will be reduced. Operating life means brightness goes down to 50% initial brightness. Typical operating life time is estimated data.

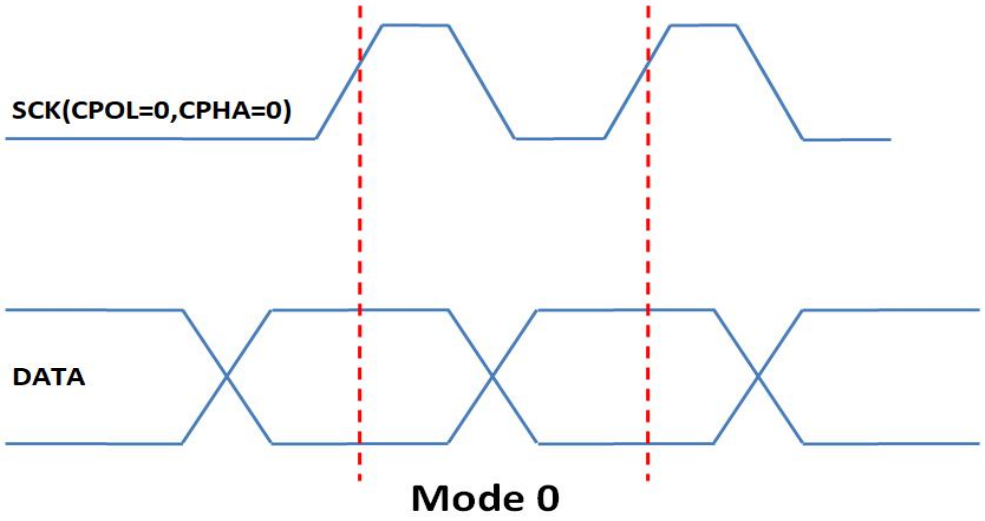
Note4: The LED driving condition is defined for each LED module.

Note5:The product has already integrated the backlight driving circuit.



5 Interface Description

If there is no special indication, TR230S works on Mode 0, That means data transmission occurs during rising edge of the clock.



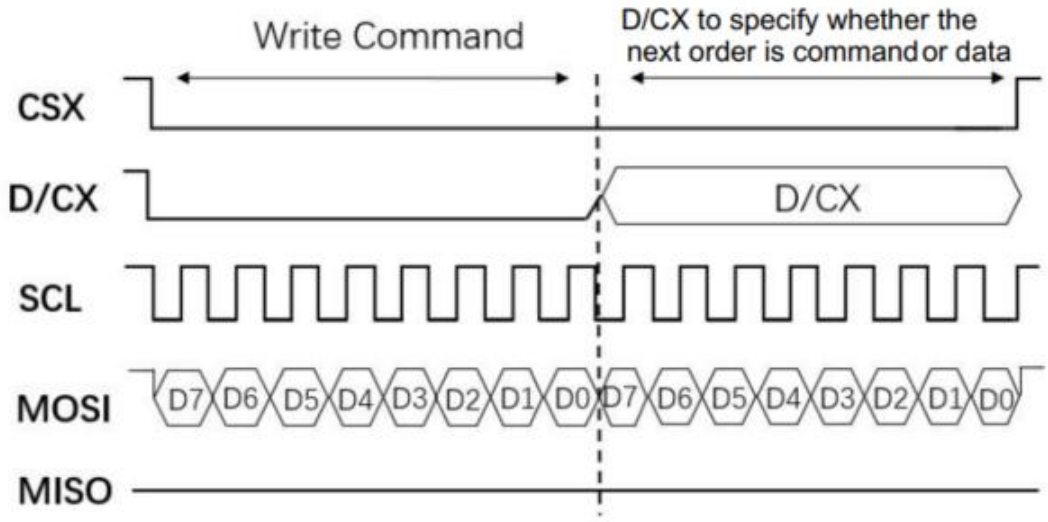
5.1 PIN DESCRIPTION

PIN NAME	Description
CS#	Chip selection: LOW active
SCL	Serial Interface Clock
SDO0	SPI:MOSI QSPI: QSPI-DATA0
SDO1	SPI:MISO QSPI: QSPI-DATA1
SDO2	QSPI: QSPI-DATA2
SDO3	QSPI: QSPI-DATA3
D/C#	DATA or COMMAND flag, 0:COMMAND 1:DATA
WAIT#	WAIT# STATUS flag, 0: Sending Command Not Allowed 1: Sending Command Allowed

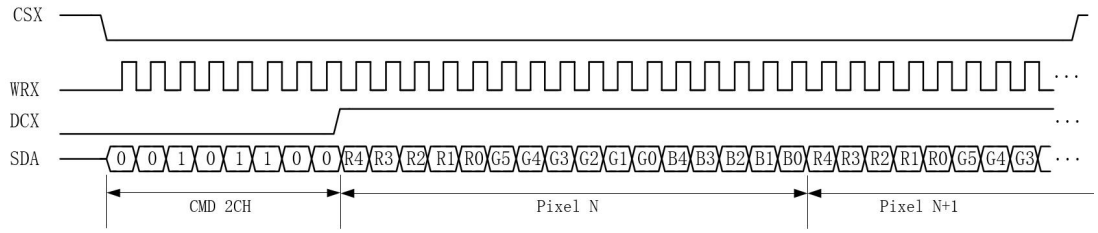
5.2 INTERFACE TIMING

- SPI-4WIRE

Timing for sending command and data:

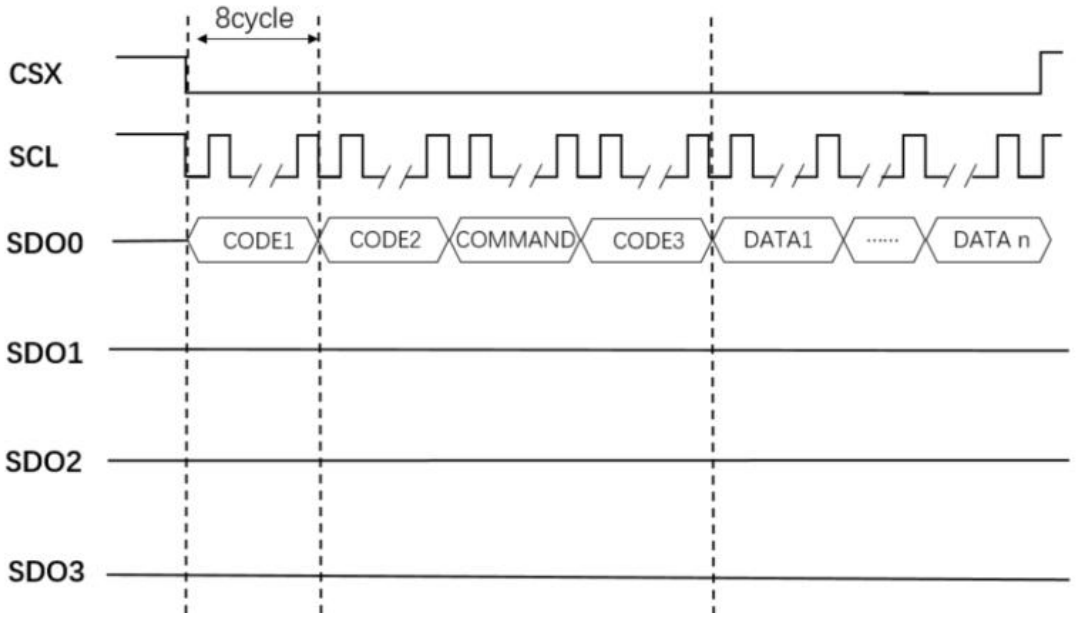


Display data writing format:

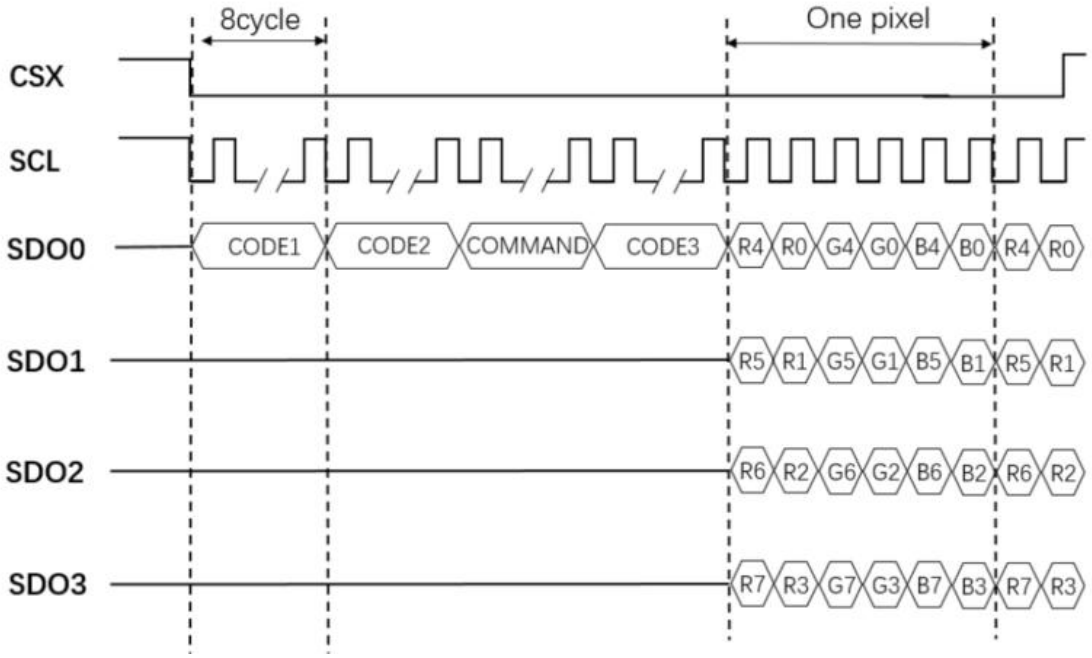


- QSPI-4SDA

Timing for sending command:
(CODE1=0x02, CODE2=CODE3=0x00)



Display data writing format:
(CODE1=0x12, CODE2=CODE3=0x00 COMMAND = 0x2C)



6 Optical Characteristics

Ta=25℃

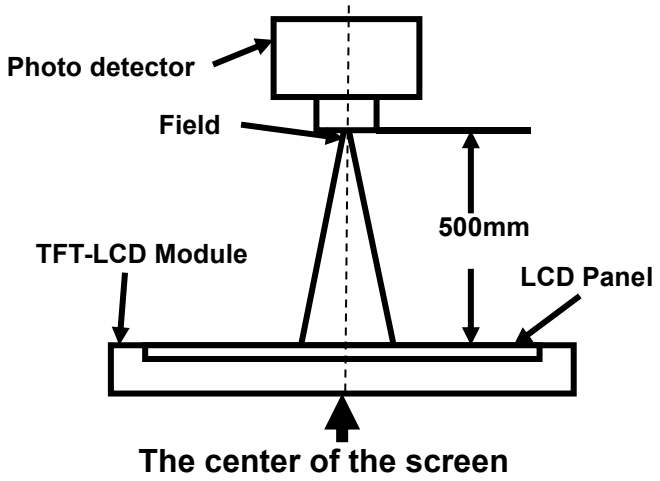
Item		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
View Angles		θT	CR≧10	80	85	-	Degree	Note 2
		θB		80	85	-		
		θL		80	85	-		
		θR		80	85	-		
Contrast Ratio		CR	θ=0°	1000	1200	-		Note1 Note3
Response Time		T_{ON}	25℃	-	30	40	ms	Note1 Note4
		T_{OFF}						
Chromaticity	White	x	Backlight is on	0.261	0.291	0.321		Note1 Note5
		y		0.261	0.291	0.321		
	Red	x		0.605	0.635	0.665		
		y		0.309	0.339	0.369		
	Green	x		0.286	0.316	0.346		
		y		0.566	0.593	0.623		
	Blue	x		0.121	0.151	0.181		
		y		0.016	0.046	0.076		
Uniformity		U		75	80	-	%	Note1 Note6
NTSC				65	70	-	%	Note 5
Luminance		L		--	585	-	cd/m^2	Note1 Note7

Test Conditions:

1. I_F= 40 mA, V_F=12.8V and the ambient temperature is 25±2℃.humidity is 65±7%
2. The test systems refer to Note 1 and Note 2.

Note 1: Definition of optical measurement system.

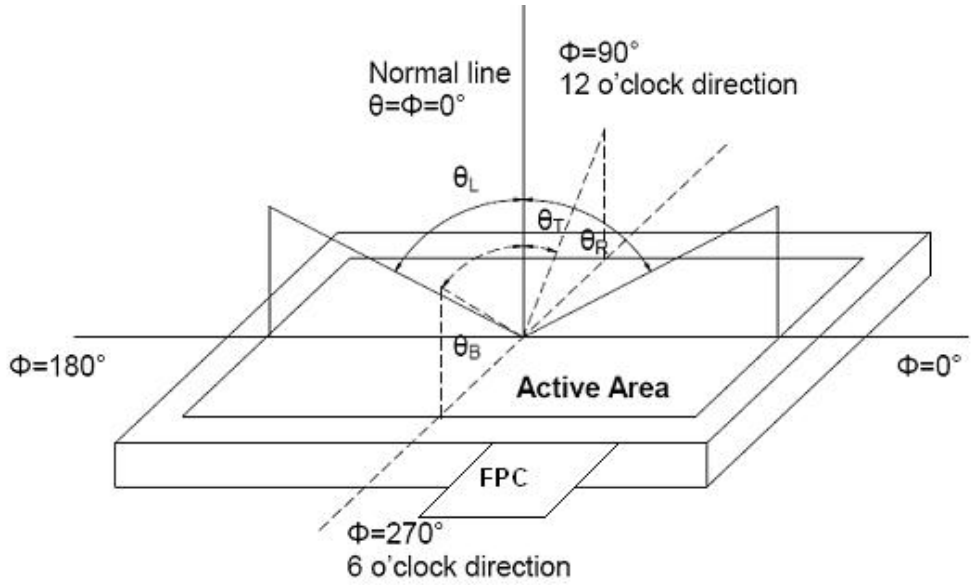
Properties are measured at the center point of the LCD screen. All input terminals LCD panel must be ground when measuring the center area of the panel.



Item	Photo detector	Field
Contrast Ratio	SR-3A	1°
Luminance		
Chromaticity		
Lum Uniformity		
Response Time	BM-7A	2°

Note 2: Definition of viewing angle range and measurement system.

Viewing angle is measured at the center point of the LCD by CONOSCOPE(ergo-80)。



Note 3: Definition of contrast ratio

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD is on the "White" state}}{\text{Luminance measured when LCD is on the "Black" state}}$$

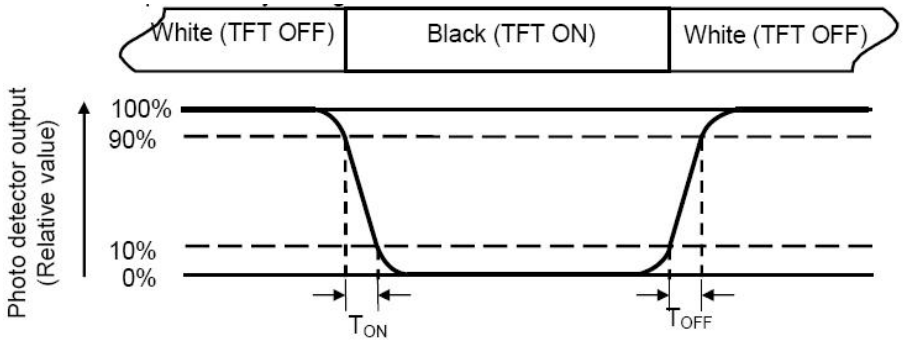
“White state “: The state is that the LCD should drive by Vwhite.

“Black state”: The state is that the LCD should drive by Vblack.

Vwhite: To be determined Vblack: To be determined.

Note 4: Definition of response time

The response time is defined as the LCD optical switching time interval between “White” state and “Black” state. Rise time (T_{ON}) is the time between photo detector output intensity changed from 90% to 10%. And fall time (T_{OFF}) is the time between photo detector output intensity changed from 10% to 90%.



Note 5: Definition of color chromaticity (CIE1931)

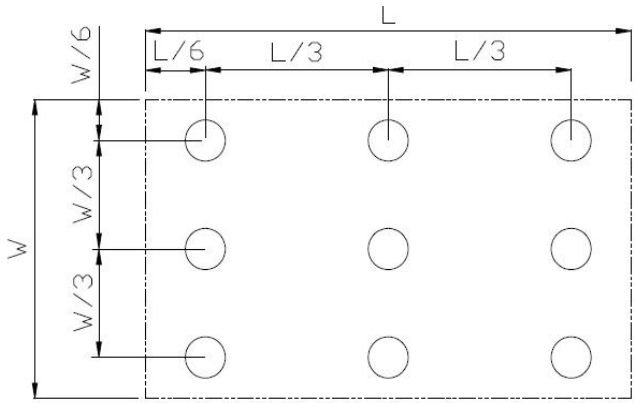
Color coordinates measured at center point of LCD.

Note 6: Definition of luminance uniformity

Active area is divided into 9 measuring areas (Refer Fig. 2). Every measuring point is placed at the center of each measuring area.

Luminance Uniformity (U) = Lmin/ Lmax

L-------Active area length W----- Active area width



Lmax: The measured Maximum luminance of all measurement position.

Lmin: The measured Minimum luminance of all measurement position.

Note 7: Definition of luminance:

Measure the luminance of white state at center point.

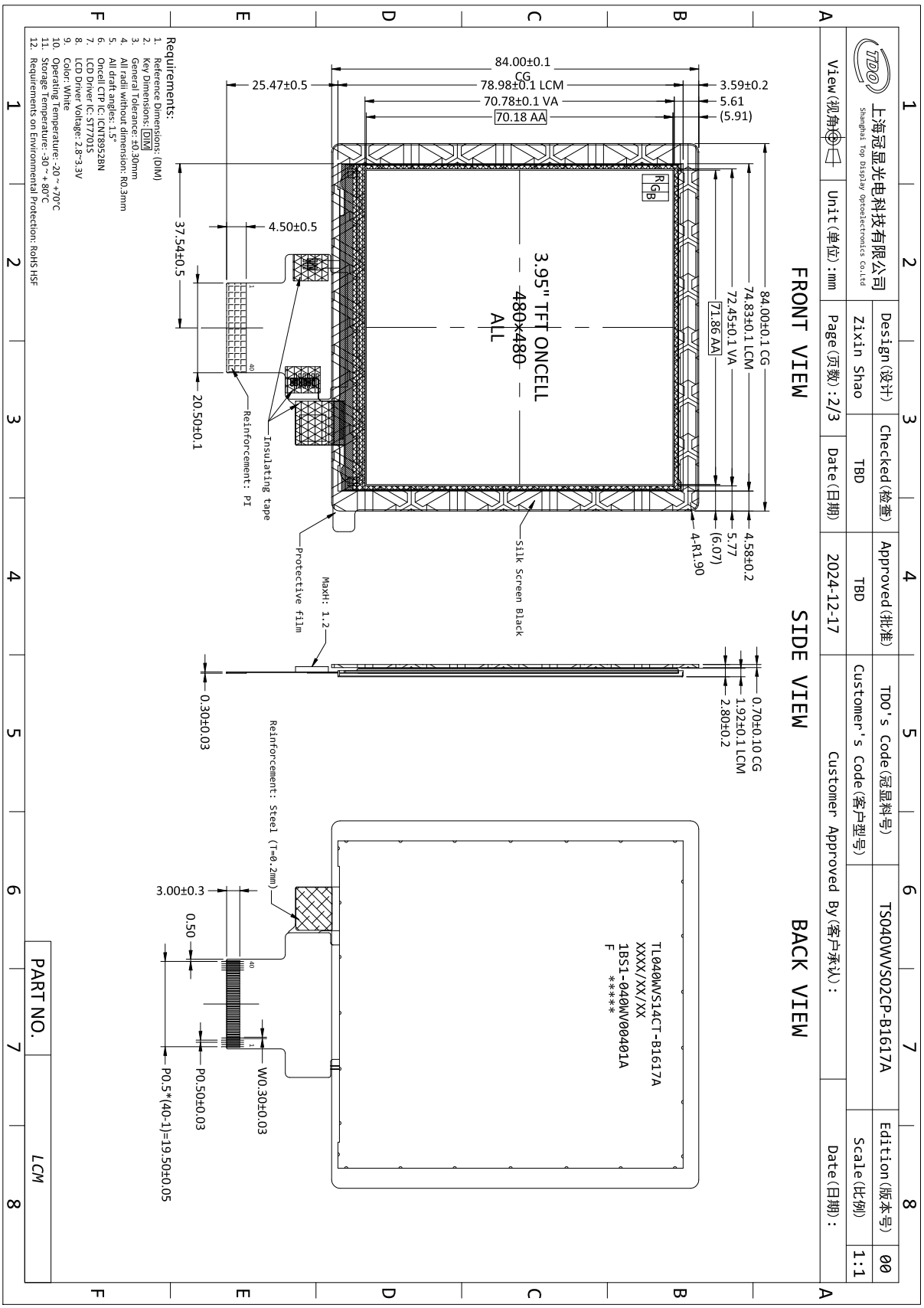
7 Environmental / Reliability Test

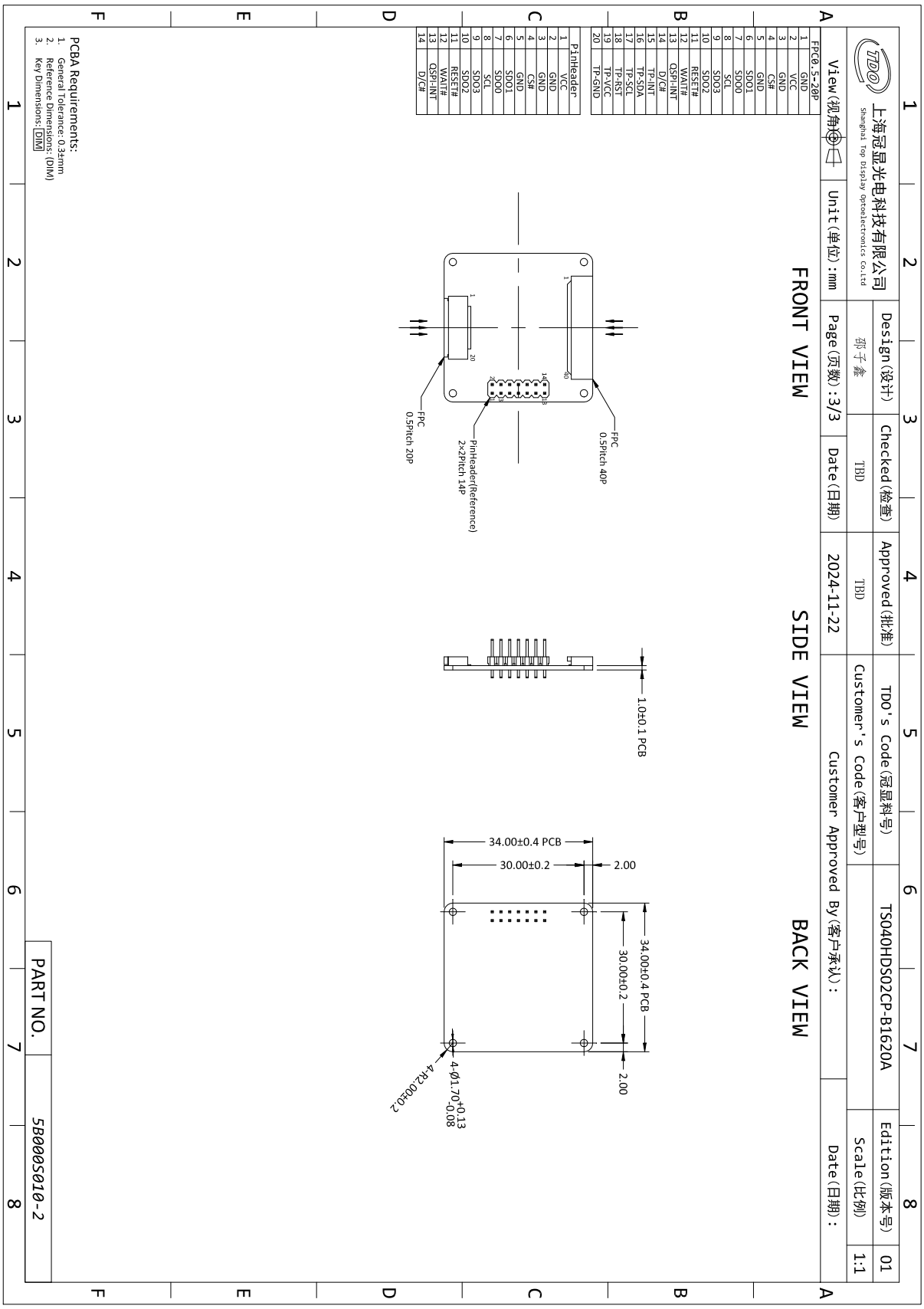
No	Test Item	Condition	Remarks
1	High Temperature Operation	Ts = +70℃, 240 hours	No abnormalities in functions
2	Low Temperature Operation	Ta = -20℃, 240 hours	No abnormalities in functions
3	High Temperature Storage	Ta = +80℃, 240 hours	No abnormalities in functions
4	Low Temperature Storage	Ta = -30℃, 240 hours	No abnormalities in functions
5	Storage at High Temperature and Humidity	Ta = +60℃, 90% RH max,240hours	No abnormalities in functions
6	Thermal Shock (non-operating)	-30℃ 30 min~ +70℃ 30 min, Change time: 0.5 hour ← 5 min → 0.5 hour.10 Cycle	Start with cold temperature, End with high temperature,
7	ESD	C=150pF, R=330Ω,5point/panel Air: ±8Kv, 5times; Contact:±4Kv,5times (Environment:15℃~35℃, 30%~60%.86Kpa~106Kpa)	No abnormalities in functions

Note1: Ts is the temperature of panel's surface.

Note2: Ta is the ambient temperature of samples.

8 Mechanical Drawing




上海冠显光电科技有限公司
Shanghai Top Display Optoelectronics Co.Ltd
Design(设计)
邵子鑫
Checked(检查)
TBD
Approved(批准)
TBD
TDO's Code(冠显料号)
TS040HDS02CP-B1620A
Edition(版本号)
01
View(视角)
Unit(单位):mm
Page(页数):3/3
Date(日期)
2024-11-22
Customer's Code(客户型号)
Customer Approved By(客户承认):
Date(日期):
Scale(比例)
1:1
FPC0.5-20P
1 GND
2 VCC
3 GND
4 CS#
5 GND
6 SDO1
7 SDO0
8 SCL
9 SDO3
10 SDO2
11 RESET#
12 WAIT#
13 QSPI-INT
14 D/C#
15 TP-INT
16 TP-SDA
17 TP-SCL
18 TP-RST
19 TP-VCC
20 TP-GND
PinHeader
1 VCC
2 GND
3 GND
4 CS#
5 GND
6 SDO1
7 SDO0
8 SCL
9 SDO3
10 SDO2
11 RESET#
12 WAIT#
13 QSPI-INT
14 D/C#
FRONT VIEW
FPC 0.5Pitch 20P
PinHeader(Reference) 2×2Pitch 14P
FPC 0.5Pitch 40P
SIDE VIEW
1.0±0.1 PCB
BACK VIEW
34.00±0.4 PCB
30.00±0.2
2.00
4-R2.00±0.2
4-Ø1.70+0.13 -0.08
PCBA Requirements:
1. General Tolerance: 0.3±mm
2. Reference Dimensions: (DIM)
3. Key Dimensions: DIM
PART NO.
5B0005010-2

9 Precautions for Use of LCD Modules

Handling Precautions

9.1.1 The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

9.1.2 If the display panel is damaged and the liquid crystal substance inside it leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

9.1.3 Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.

9.1.4 The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully.

9.1.5 If the display surface is contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If still not completely clear, moisten cloth with one of the following solvents:

- Isopropyl alcohol
- Ethyl alcohol

Solvents other than those mentioned above may damage the polarizer. Especially, do not use the following:

- Water
- Ketone
- Aromatic solvents

9.1.6 Do not attempt to disassemble the LCD Module.

9.1.7 If the logic circuit power is off, do not apply the input signals.

9.1.8 To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.

9.1.8.1 Be sure to ground the body when handling the LCD Modules.

9.1.8.2 Tools required for assembly, such as soldering irons, must be properly ground.

9.1.8.3 To reduce the amount of static electricity generated, do not conduct assembly and other work under dry conditions.

9.1.8.4 The LCD Module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

Storage Precautions

9.2.1 When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

9.2.2 The LCD modules should be stored under the storage temperature range. If the LCD modules will be stored for a long time, the recommend condition is: Temperature : 0℃ ～ 40℃ Relatively humidity: ≤80%

9.2.3 The LCD modules should be stored in the room without acid, alkali and harmful gas.

Transportation Precautions

9.3.1 The LCD modules should be no falling and violent shocking during transportation, and also should avoid excessive press, water, damp and sunshine.